

MATH Unit 1 — Matrices & Linear Algebra

Complete Notes

Mathematics (MATH) · Unit 1 · Matrices & Linear Algebra · 2024

Overview

These notes cover **Matrices & Linear Algebra**. Engineering and discrete mathematics — linear algebra, calculus, probability, logic and graph theory for computer science. Each section states the core idea, a worked intuition and the points examiners look for.

1. Rank of a Matrix

The rank is the number of linearly independent rows, found by reducing to echelon form.

Why it matters: Rank of a Matrix appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of rank of a matrix in one line.
- A small example showing rank of a matrix in practice.
- Common mistakes students make and how to avoid them.

2. Systems of Linear Equations

Consistency of $AX = B$ is decided by comparing the ranks of A and the augmented matrix.

Why it matters: Systems of Linear Equations appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of systems of linear equations in one line.
- A small example showing systems of linear equations in practice.
- Common mistakes students make and how to avoid them.

3. Eigenvalues and Eigenvectors

Eigenvalues solve $\det(A - \lambda I) = 0$; eigenvectors span the corresponding null spaces.

Why it matters: Eigenvalues and Eigenvectors appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of eigenvalues and eigenvectors in one line.
- A small example showing eigenvalues and eigenvectors in practice.
- Common mistakes students make and how to avoid them.

4. Cayley-Hamilton Theorem

Every square matrix satisfies its own characteristic equation, which helps compute inverses and powers.

Why it matters: Cayley-Hamilton Theorem appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of cayley-hamilton theorem in one line.
- A small example showing cayley-hamilton theorem in practice.
- Common mistakes students make and how to avoid them.

5. Diagonalization

A matrix with n independent eigenvectors can be written as $P D P^{-1}$.

Why it matters: Diagonalization appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of diagonalization in one line.
- A small example showing diagonalization in practice.
- Common mistakes students make and how to avoid them.

6. Quadratic Forms

Quadratic forms are classified as positive definite, negative definite or indefinite using eigenvalues.

Why it matters: Quadratic Forms appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of quadratic forms in one line.
- A small example showing quadratic forms in practice.
- Common mistakes students make and how to avoid them.

Quick revision

Rank of a Matrix · Systems of Linear Equations · Eigenvalues and Eigenvectors · Cayley-Hamilton Theorem · Diagonalization · Quadratic Forms